Nathaniel Tornow

PhD student at Technical University of Munich

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Research Interests

My research interests lie at the intersection of compiler design, software systems, and fault-tolerant quantum computing, with a focus on algorithm-architecture co-design.

My research addresses a fundamental gap in fault-tolerant quantum computing: we cannot determine which algorithms are viable on real hardware, because no end-to-end algorithm-aware compiler exists to verify resource estimates. I address this through automated synthesis and structured compilation to enable verifiable, scalable algorithm-architecture co-design.

Education

Ph.D. in Computer Science (2025 — present)
Technical University of Munich
Thesis: Verifiable Co-Design of Fault-Tolerant Quantum Algorithms and Architectures
Advisor: Prof. Dr. Pramod Bhatotia

M.Sc. in Computer Science (2022 — 2025)
Technical University of Munich
Thesis: Scalable Hybrid Quantum-Classical Computing with Tensor Networks (Grade: 1.0)

B.Sc. in Computer Science (2018 — 2022)
Technical University of Munich
Thesis: Efficient Distributed Simulation of Large Quantum Circuits (Grade: 1.0)

Publications

1. AtomGuard: A Neutral Atom Quantum ISA for Compositionally Verified Hardware Safety
   *Francisco Romao, **Nathaniel Tornow**, Emmanouil Giortamis, Dennis Sprokholt, Pramod Bhatotia*
   ACM ASPLOS 2027 (under submission)

2. qTPU: Hybrid Tensor Networks for Quantum-Classical Acceleration
   ***Nathaniel Tornow**, Emmanouil Giortamis, Dennis Sprokholt, Christian Mendl, Pramod Bhatotia*
   USENIX OSDI 2026 (accepted)

3. Qonductor: A Cloud Orchestrator for Quantum Computing
   *Emmanouil Giortamis, Francisco Romao, **Nathaniel Tornow**, Dmitry Lugovoy, Pramod Bhatotia*
   ACM SC 2025
   ⋆ **Best Student Paper Award Finalist**

4. QOS: Quantum Operating System
   *Emmanouil Giortamis, Francisco Romao, **Nathaniel Tornow**, Pramod Bhatotia*
   USENIX OSDI 2025

5. QVM: Quantum Gate Virtualization Machine
   ***Nathaniel Tornow**, Emmanouil Giortamis, Pramod Bhatotia*
   ACM PLDI 2025

6. FlexLog: A Shared Log for Stateful Serverless Computing
   *Dimitra Giantsidi, Emmanouil Giortamis, **Nathaniel Tornow**, Florin Dinu, Pramod Bhatotia*
   ACM HPDC 2023

Honors and Awards

Google PhD Fellowship — Nominated by TU Munich (1 of 3, 2026)

Best Student Paper Award Finalist, ACM SC'25 (Qonductor)

1st prize (Optiver challenge), HackZurich Hackathon (2021)

2nd prize (Microsoft challenge), HackaTUM Hackathon (2020)

Employment

QuEra Computing, Sep — Nov 2026
Research Intern
Compiler research for neutral-atom fault-tolerant quantum computing.

planqc, Apr — Oct 2025
Software Engineer Intern
Led the development of compiler infrastructure for neutral-atom quantum processors.

Walther-Meissner-Institut, Oct 2024 — Apr 2025
Research Project
Developed automatic calibration software for superconducting qubit systems.

Leibniz Supercomputing Centre, Oct 2022 — Mar 2025
Research Assistant
Developed compiler passes for the Munich Quantum Software Stack. Contributed to HPC-quantum integration infrastructure connecting classical supercomputers with diverse QPUs.

Teaching

Introduction to Software Engineering (EIST): Spring 2026

Discrete Probability Theory: Fall 2021, Fall 2022

Distributed Systems: Spring 2021, Spring 2022

Practical Lab: Systems Programming: Fall 2022

Practical Lab: Introduction to Programming: Fall 2019

Supervised Theses

1. Co-advising an ongoing MSc thesis on magic state cultivation scheduling (2025–2026)
2. CompTN: A Compiler Infrastructure for High-Performance Tensor Network Computing
   Francisco Kush, BSc Thesis, 2024

Service

External Reviewer: ASPLOS'24, PLDI'26

Conference Student Volunteer: IEEE Quantum Week 2024

References

Prof. Dr. Pramod Bhatotia
Full Professor, Systems Research Group
Technical University of Munich
`bhatotia@in.tum.de`

Prof. Dr. Christian Mendl
Associate Professor, Quantum Computing
Technical University of Munich
`christian.mendl@tum.de`

Adrian Vetter
Head of Software
planqc GmbH
`adrian@planqc.eu`